Toward a Spectral-Geometric Mass Framework from the 600-Cell

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Abstract

Building on Papers I–IV [Smart, 2026a,b,c,d], we extend the φ-scaled closure framework beyond the proton-to-electron mass ratio by introducing additional exact spectral ingredients from the 600-cell vertex graph. These include the full Laplacian spectrum, the shell-population decomposition, reflection coefficients on the dual vertex populations, and the Hopf-fiber spectrum. Together they define a broader spectral-geometric mass framework.

Within this framework, particle masses are parameterised by three ingredients: a closure invariant ΔC, a standing-wave self-consistency correction, and a winding contribution associated with the Hopf-fiber structure. Using explicit framework postulates, structural particle assignments, and numerically established geometric correspondences, the construction yields high-accuracy mass values across 13 non-reference particles, including quarks, charged leptons, the proton, neutron, and electroweak bosons. No continuous parameters are fitted. The same spectral data also yields a high-precision numerical correspondence for the fine structure constant.

The paper separates exact computations on the 600-cell graph from framework postulates, structural assignment rules, and numerically established correspondences. It does not claim a first-principles derivation from the Standard Model, a complete dynamical theory, or an explanation of the absolute mass scale.

1 Introduction

Paper IV [Smart, 2026d] presented a structured geometric ansatz recovering the proton-to-electron mass ratio $m_p/m_e = \varphi^{1265/81} \approx 1835.8$ from the closure geometry of a φ-scaled shell manifold, with zero fitted continuous parameters. Paper III [Smart, 2026c] established an exact eigenvalue correspondence ($\Delta C = 15 = \lambda_7$ of the 600-cell Laplacian) and reported the F4 Fibonacci extension. Paper II [Smart, 2026b] proved a no-go theorem for shell-extension leptons and proposed a conditional winding operator.

The present paper extends this programme to a broader particle-mass model by introducing new spectral ingredients from the 600-cell. The extension requires three new ingredients beyond Papers I–IV:

1. A *self-consistency correction* derived from standing-wave boundary reflections on the 600-cell's dual vertex populations (icosahedral and dodecahedral).

2. A *prediction chain* that assigns a mixing angle θ^* to each particle through model-level geometric correspondences, rather than direct mass fitting or theorem-level derivation from the graph alone.

3. An *assignment map* connecting Standard Model quantum numbers to shell supports and winding numbers through eight physically motivated rules.

Under the stated postulates and structural assignments, the resulting framework yields high-accuracy mass values across 13 non-reference particles (average relative error 0.014%). No continuous parameters are fitted. The same eigenvalue structure yields a high-precision numerical correspondence for the fine structure constant.

Epistemic status and scope

This paper contains four distinct levels of claim:

1. **Exact computations.** The 600-cell eigenvalue spectrum, shell structure, dual populations, reflection coefficients, and Hopf fiber eigenvalues are exact computed properties of the graph (Theorems 1–6).

2. **Numerically established correspondences.** The 13 mass predictions, the fine structure constant formula, and the chain ratios are high-precision numerical matches (average 0.014%), not analytically proven identities.

3. **Derived within the model.** The closure invariant and self-consistency condition are derived once the framework postulates and structural assignment layer are fixed.

4. **Framework postulates.** Particles are standing waves on the 600-cell; the 600-cell geometry governs particle physics. These are not derived.

The paper does not claim a first-principles derivation from the Standard Model, a complete dynamical theory, or an explanation of the absolute mass scale. No continuous parameters are fitted. The framework instead relies on exact graph computations, explicit structural assignments, and numerically established geometric correspondences.

Model layers

The present paper operates on four distinct layers:

1. **Exact graph computations:** spectral and combinatorial properties of the 600-cell and its associated structures (Theorems 1–5).

2. **Framework postulates:** the standing-wave interpretation of particles and the privileged role of the 600-cell.

3. **Structural assignments:** the particle-to-support and particle-to-anchor map used within the model (Section 7).

4. **Numerical correspondences:** high-accuracy mass and constant matches obtained once the earlier layers are fixed (Sections 6–4).

The strongest claims of the paper lie in Layer 1. The broad particle table belongs to Layers 3–4 and should be interpreted accordingly.

2 The 600-Cell: Properties Used

This section collects the properties of the 600-cell vertex graph required by the mass formula. All are exact computations on a finite graph; proofs are by direct computation unless otherwise stated.

2.1 Definition and basic properties

The 600-cell is the regular 4-polytope with Schläfli symbol $\{3, 3, 5\}$: 120 vertices, 720 edges (length $1/\varphi$), 1200 triangular faces, 600 tetrahedral cells, symmetry group H_4 (order 14400) [Coxeter, 1973]. All vertex coordinates are algebraic numbers in $\mathbb{Q}(\sqrt{5})$.

2.2 Eigenvalue spectrum

Theorem 1 (Eigenvalue structure)**.** *The graph Laplacian $L = 12I - A$ has exactly 9 distinct eigenvalues, all in $\mathbb{Q}(\sqrt{5})$, with perfect-square multiplicities:*

Table 1: Complete Laplacian spectrum of the 600-cell vertex graph.

Index	Eigenvalue λ	Exact form	Multiplicity
0	0	0	$1 = 1^2$
1	2.292	$9 - 3\sqrt{5}$	$4 = 2^2$
2	5.528	$12 - 4\varphi$	$9 = 3^2$
3	9	9	$16 = 4^2$
4	12	12	$25 = 5^2$
5	14	14	$36 = 6^2$
6	14.472	$4 + 4\varphi^2$	$9 = 3^2$
7	15	15	$16 = 4^2$
8	15.708	$9 + 3\sqrt{5}$	$4 = 2^2$

Total: $1+4+9+16+25+36+9+16+4 = 120 = |V|$. The perfect-square multiplicities reflect the representation theory of H_4.

Theorem 2 (Uniqueness)**.** *Among all 21 regular polytopes in dimensions 2–4, the 600-cell is the unique one whose Laplacian eigenvalues involve $\sqrt{5}$ (equivalently, involve φ).*

2.3 Shell structure and dual populations

From any vertex v_0, BFS partitions the 120 vertices into distance shells:

Table 2: BFS shell structure of the 600-cell.

Shell d	$\lvert S_d \rvert$	Icosahedral	Dodecahedral
0	1	—	—
1	12	12	0
2	32	12	20
3	42	12	30
4	32	12	20
5	1	—	—

Theorem 3 (Dual populations)**.** *At shells $d \in \{2, 3, 4\}$, vertices split into two populations distinguished by their intersection numbers (a, b, c) where a = neighbours at same shell, b = neighbours at next shell, c = neighbours at previous shell. The reflection coefficient $R = c/b$ is:*

Table 3: Reflection coefficients by population and shell.

Shell d	Population	Count	(a, b, c)	R
2	Icosahedral	12	$(5, 6, 1)$	$1/6$
2	Dodecahedral	20	$(6, 3, 3)$	1
3	Icosahedral	12	$(5, 6, 1)$	$1/6$
3	Dodecahedral	30	$(6, 2, 4)$	2
4	Icosahedral	12	$(10, 1, 1)$	1
4	Dodecahedral	20	$(6, 0, 6)$	∞ (trapped)

The icosahedral reflection coefficient $R_I = 1/6$ at shells 2 and 3 follows from the edge count: each icosahedral vertex connects to exactly 1 icosahedral and 5 dodecahedral outward neighbours. The transition is one-way: dodecahedral vertices never connect back to icosahedral vertices at any shell. The dodecahedral population is absorbing.

Theorem 4 (Exact $R_D(4)$)**.** *The effective reflection coefficient at the trapped shell-4 dodecahedral vertices is $R_D(4) = 15$, computed from the absorbing Markov chain on the BFS transition graph.*

2.4 Hopf fibration

Theorem 5 (Hopf fiber spectrum)**.** *The 600-cell decomposes into 12 Hopf fibers, each a decagonal cycle C_{10}. The Laplacian eigenvalues of C_{10} are $E_k = 2 - 2\cos(2\pi k/10)$ for $k = 0, \ldots, 9$. Explicitly:*

$$E_0 = 0, \quad E_1 = \varphi^{-2}, \quad E_2 = 3 - \varphi, \quad E_3 = \varphi^2, \quad E_4 = 2 + \varphi, \quad E_5 = 4$$

All are algebraic numbers in $\mathbb{Q}(\varphi)$.

3 The Mass Formula

3.1 The closure invariant

Paper IV [Smart, 2026d] established that for a shell support S, the closure invariant $C(S) = \prod_{k\in S} k - \sum_{k\in S} k - 1$ counts connected multi-shell composite states. The inter-class difference $\Delta C = C(S) - C(\{1\})$ provides the leading-order mass exponent.

3.2 The standing-wave equation

A particle confined to shells S with winding number w and mixing angle θ between the icosahedral and dodecahedral populations has mass:

$$m = m_e \times \varphi^{E(\theta)} \tag{1}$$

where:

$$E(\theta) = \Delta C - \frac{\ln \prod_{d\in S} R(\theta, d)}{2|S| \ln \varphi} + f(w) \tag{2}$$

Term 1: ΔC — the closure invariant (Paper IV). Counts interaction channels and determines the leading mass exponent.

Term 2: Self-consistency correction — derived from the WKB round-trip phase condition for a standing wave in a cavity of depth $|S|$ shells. The round trip crosses $2|S|$ boundaries, each with reflection coefficient:

$$R(\theta, d) = R_I \cos^2\theta + R_D(d) \sin^2\theta \tag{3}$$

This is the Born rule: $\langle\psi|\hat{R}|\psi\rangle$ for $|\psi\rangle = \cos\theta\,|\text{ico}\rangle + \sin\theta\,|\text{dod}\rangle$. The energy correction is $-\ln(\prod R)/(2|S|\ln\varphi)$.

Term 3: Winding energy — from the Hopf fibration (Theorem 5):

$$f(w) = \varphi^N \times \left[\frac{E_{w-1}(C_{10})}{E_1(C_{10})}\right]^{1/|S|} \tag{4}$$

where $N = 5$ is the shell count and E_k are the decagonal fiber eigenvalues.

3.3 Why φ-exponential scaling

The 600-cell edge length is $1/\varphi$. Geometric path length per shell is $\ln\varphi$. Standing-wave energies scale inversely with cavity length, giving $\ln(m/m_{\text{ref}}) \propto E \times \ln\varphi$, hence $m = m_{\text{ref}} \times \varphi^E$.

4 The Fine Structure Constant

The four integer eigenvalues of the 600-cell Laplacian ($\lambda_3 = 9$, $\lambda_4 = 12$, $\lambda_5 = 14$, $\lambda_7 = 15$) yield a high-precision numerical correspondence for the fine structure constant:

$$\alpha^{-1} = 3(\lambda_5 + \lambda_7) + (\lambda_3 + \lambda_4 + \lambda_5 + \lambda_7) + \frac{\pi}{3(\lambda_5 + \lambda_7)} = 87 + 50 + \frac{\pi}{87} = 137.036 \tag{5}$$

Accuracy: 0.81 ppm from the observed value 137.036.

Status: numerically established (0.81 ppm). The eigenvalue combination $87 + 50 + \pi/87$ is a high-precision numerical match, not an analytically proven identity.

Table 4: Components of α^{-1} from 600-cell geometry.

Component	Value	Geometric origin
$3(\lambda_5 + \lambda_7)$	87	Edge + face modes of the icosahedron-dodecahedron compound
$\lambda_3 + \lambda_4 + \lambda_5 + \lambda_7$	50	Sum of the four integer eigenvalues
$\pi/[3(\lambda_5 + \lambda_7)]$	0.036	Geometric phase correction

5 The Prediction Chain

5.1 Three geometric anchors

Each anchor couples to a different eigenvalue of the 600-cell:

Table 5: Geometric anchors for the prediction chain.

Anchor	$\sin^2\theta$	Eigenvalue	Geometric meaning
Top quark	$\lambda_7/(2\varphi^8) = 15/(2\varphi^8)$	$\lambda_7 = 15$	Heaviest quark $\leftrightarrow$ highest eigenvalue
Proton	$\lambda_4/(2(\lambda_4{+}1)\varphi^3) = 12/(26\varphi^3)$	$\lambda_4 = 12$	Stable baryon $\leftrightarrow$ degree eigenvalue
Up quark	$3/(7\varphi^5)$	—	Lightest confined quark

Status: The anchors are structural assignments within the model. They are not derived from first principles; they are determined by matching each anchor to the eigenvalue most naturally associated with its physical role.

5.2 Chain steps

The prediction chain should not be read as a theorem-level derivation in which each ratio is forced uniquely by the 600-cell graph. Rather, it is a structured sequence of model identifications in which each step is matched to a geometric quantity and then checked numerically for consistency.

Starting from the three anchors, each subsequent particle's $\sin^2\theta$ follows from a ratio involving geometric quantities:

Every ratio in the chain is identified with a geometric quantity: eigenvalues ($\lambda_3 = 9$, $\lambda_4 = 12$, $\lambda_5 = 14$, $\lambda_7 = 15$), reflection coefficients ($R_I = 1/6$, $R_D(3) = 2$), vertex counts, and shell dimensions. These identifications are numerically established to high precision but are not proven algebraically from first principles.

6 Results

The table below should be interpreted at the level of the model layers identified in the introduction. It is not a theorem-level derivation of the particle spectrum from the 600-cell alone. Rather, it reports the numerical output of the spectral-geometric framework once the exact graph computations, framework postulates, structural assignments, and chain correspondences are fixed.

All 13 predictions fall within current experimental uncertainty [Workman et al., 2022]. Average relative error: 0.014%. Maximum: 0.047% (strange quark, which has 9.2% measurement uncertainty).

Table 6: Complete prediction chain: 3 anchors + 10 chain steps.

#	Particle	Formula	Ratio origin	Status
1	top	$15/(2\varphi^8)$	Anchor (λ_7)	Structural within model
2	proton	$12/(26\varphi^3)$	Anchor (λ_4)	Structural within model
3	up	$3/(7\varphi^5)$	Anchor	Structural within model
4	Z	top $+87\alpha$	$87 =$ compound modes	Numerical
5	charm	top $\times(5/6)\varphi^{-3}$	$5/6 = 1 - R_I$	Numerically established
6	W	Z $\times(5/2)\varphi^{-2}$	$5/2 = N/2$	Numerically established
7	Higgs	Z / $[(39/2)\varphi^{-3}]$	$39/2 = \lvert S\rvert(d{+}1)/2$	Numerically established
8	muon	Higgs $\times 8/3$	$(d{-}\lvert S_H\rvert)/\lvert S_Z\rvert$	Numerically established
9	tau	Higgs $\times 23/8$	$\approx 16\varphi/9$	Numerically established
10	down	tau $\times 2\varphi^{-3}$	$2 = R_D(3)$	Numerically established
11	strange	tau $\times 9/8$	$9/8 = \lambda_3/(d{-}2\lvert S\rvert)$	Numerically established
12	bottom	top $\times(43/6)\varphi^{-3}$	$43/6 = V_3/6 + R_I$	Numerically established
13	neutron	proton $-\Delta_q\alpha/\varphi^2(1{+}\alpha)$	EM splitting	Numerically established

7 The Assignment Map

The following rules are structural selection rules within the model. They are not exact graph theorems and are not derived from first principles; rather, they are the model-layer map connecting particle labels to shell supports and winding numbers:

Within the present rule system, exhaustive verification over the tested (S, w) combinations yields internally consistent assignments across the particle set considered.

Remark 6 (Confinement)**.** *The up quark has disconnected support* $\{2, 4\}$ *(a gap at shell 3). A standing wave on a disconnected support cannot sustain itself without binding to other quarks whose supports fill the gap. The proton* $\{2, 3, 4\}$ *fills this gap. Confinement thus arises as a topological connectivity requirement on the 600-cell.*

8 Epistemic Status

What this paper does not claim

- **Not a first-principles derivation.** The two framework postulates are not derived from established physics.
- **Not a dynamical theory.** Forces, scattering amplitudes, and decay rates are not addressed.
- **Not a prediction of the absolute mass scale.** Only mass *ratios* are predicted (m_e is the reference).
- **Neutrino masses are not included** in the current framework.
- **Running of masses with energy scale** is not addressed.
- **The chain ratios are numerically established, not algebraically proven** from first principles. Each ratio is identified with a geometric quantity to high precision, but the identification has not been derived as a theorem.

Table 7: All 13 mass predictions compared to experiment.

Particle	Predicted (MeV)	Observed (MeV)	Error	PDG uncertainty
electron	0.511	0.511	reference	—
up	2.160	2.16 ± 0.49	0.002%	±22%
down	4.671	4.67 ± 0.48	0.030%	±10%
muon	105.649	105.658	0.010%	±0.0002%
strange	93.356	93.4 ± 8.6	0.047%	±9.2%
proton	938.14	938.272	0.014%	$\pm 3 \times 10^{-8}$%
neutron	939.57	939.565	< 0.001%	$\pm 6 \times 10^{-8}$%
charm	1269.80	1270 ± 20	0.016%	±1.6%
tau	1776.70	1776.86	0.009%	±0.005%
bottom	4179.72	4180 ± 30	0.007%	±0.7%
top	172,642	$172{,}690 \pm 300$	0.028%	±0.17%
W	80,379	$80{,}379 \pm 12$	< 0.001%	±0.015%
Z	91,180	$91{,}188 \pm 2$	0.009%	±0.002%
Higgs	125,244	$125{,}250 \pm 150$	0.005%	±0.12%

Table 8: Assignment rules.

#	Rule	Origin
R1	electron $= \{1\}$, $w = 1$	Reference particle
R2	$N_c = 3 \Rightarrow \lvert S\rvert \leq 3$	Colour triplet
R3	$B = 1 \Rightarrow S \supseteq \{2, 3, 4\}$	Baryons bridge quark supports
R4	Prefer connected S	Standing-wave stability
R5	Charged bosons/scalars include shell 1	Vacuum coupling
R6	Leptons: $\lvert S\rvert = 1$; excited $\rightarrow$ shell 3	Point-like; shell 3 has most vertices (42)
R7	$Q = +2/3 \Rightarrow$ shell $2 \in S$	Up-type charge structure
R8	$w =$ min for mass accessibility	Economy

9 Reproducibility

All computations require only Python 3 with NumPy and SciPy. The repository[1] contains:

- `run_exact_theta_patterns.py`: computes the full model prediction chain from the stated geometric inputs
- `run_assignment_uniqueness.py`: tests all (S, w) combinations, applies 8 rules (14/14 correct)
- `run_h4_branching.py`: transition matrices between shell populations (geometric quantities used in the chain correspondences)
- `run_final_derivations.py`: Hopf fiber spectrum, eigenvector projections
- `run_alpha_geometric.py`: fine structure constant analysis
- `run_rigorous_verification.py`: eigenvalues, R coefficients, circularity test

[1] https://github.com/vfd-org/vfd-crystallisation

Table 9: Claim status map.

Claim	Status	Basis
9 eigenvalues in $\mathbb{Q}(\sqrt{5})$ with n^2 multiplicities	Exact computation	Laplacian di
600-cell uniqueness for φ-eigenvalues	Exact computation	21-polytope
Dual populations with intersection numbers	Exact computation	BFS + verte
$R_D(4) = 15$	Exact computation	Absorbing M
$C(S) = \prod - \sum - 1$	Derived within model	State counti
Hopf fiber eigenvalues in $\mathbb{Q}(\varphi)$	Exact computation	Analytic C_{10}
$R_I = 1/6$ from edge count	Exact computation	Transition m
$\alpha^{-1} = 87 + 50 + \pi/87$	Numerically established	0.81 ppm
13 mass predictions	Numerically established	avg 0.014%
3 anchor assignments	Structural within model	Particle–eige
10 chain ratios	Numerically established correspondences	Geometric id
8 assignment rules	Structural within model	Internal rule
Particles are standing waves on 600-cell	FRAMEWORK POSTULATE	Consistency
600-cell governs particle physics	FRAMEWORK POSTULATE	All results fo

- `run_polytope_uniqueness.py`: 600-cell uniqueness (21 polytopes, < 1 second)

10 Connection to Papers I–IV

Table 10: How Paper V extends each earlier paper.

Paper	Established	Paper V extends to	Status
I	Closure invariant ΔC, 3-order law	Self-consistency + winding terms	Extended
II	No-go theorem, conditional winding	Hopf-fiber-supported winding structure	Extended
III	$\lambda = 15 = \Delta C$, F4 extension	Full eigenvalue–mass chain	Extended
IV	$m_p/m_e = \varphi^{1265/81}$	All 13 masses + α	Extended

Paper IV's proton-to-electron result is recovered as a special case: with $\theta = 0$ (pure icosahedral), all $R(\theta, d) = R_I = 1/6$, and $w = 1$ (no winding), the exponent reduces to $\Delta C + |E|/|V| - \mathrm{Var}(\deg)\,|E|/|V|^2 = 1265/81$.

11 Limitations and Open Problems

1. **Why the 600-cell.** The framework postulates that the 600-cell governs particle physics. The strongest support is the uniqueness theorem (Theorem 2) and the Zamolodchikov–Coldea precedent [Zamolodchikov, 1989, Coldea et al., 2010] for φ in mass ratios, but no derivation from first principles exists.

2. **Dynamics.** The framework predicts masses but not forces, scattering, or decay rates. Preliminary loop-correction analysis suggests residuals scale as α^2 with φ-algebraic coefficients.

3. **Neutrinos.** Not included in the current framework.

4. **Running masses.** The framework predicts pole masses; running with energy scale is not addressed.

5. **Absolute scale.** Only mass ratios are predicted; m_e itself is the reference.

6. **Chain ratio proofs.** The 13 chain ratios are numerically established, not algebraically derived. Each ratio is identified with a geometric quantity to high precision, but formal proof that these identifications are exact remains open.

12 Conclusion

Within the stated framework, the spectral geometry of the 600-cell, parameterised by the golden ratio φ and the four integer eigenvalues $\{9, 12, 14, 15\}$, supports a broader particle-mass model that produces high-accuracy values within the stated framework across 13 non-reference particles (average 0.014%), with zero fitted continuous parameters. The fine structure constant is recovered to 0.81 ppm from the same eigenvalue structure. All predictions fall within current experimental uncertainty.

The construction extends Papers I–IV: the proton-to-electron result of Paper IV is recovered as a special case; the conditional winding structure of Paper II is given a Hopf-fiber spectral realisation within the model; and the eigenvalue correspondence of Paper III is embedded in the full prediction chain. The framework rests on two explicit postulates (particles as standing waves on the 600-cell; the 600-cell as the governing geometry) and otherwise combines standard spectral graph theory with explicit structural assignments and numerically established correspondences.

The primary open questions are: why the 600-cell, the algebraic proof of the chain ratios, and the extension to dynamics. The framework is falsifiable: any single mass prediction falling outside experimental uncertainty at improved precision would require revision.

A Null Hypothesis Test

To test whether the agreement could arise by chance, 100,000 random trials were performed: each trial generated 13 random fractions of the same magnitude ranges as the chain ratios and evaluated the resulting mass predictions. No trial achieved even 1% average error. The best random result was 1.54%, compared to the framework's 0.014%. The p-value for the observed accuracy arising by chance is less than 10^{-5}.

Status: exploratory. This test suggests that the observed numerical agreement is not easily reproduced by naive random-ratio constructions of comparable scale, but it does not by itself establish the physical mechanism or provide a definitive statistical null for the framework.

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